Date :

7. Polymorphism

7.1 Method Overloading in Java

If a class has multiple methods having same name but different in parameters, it is known as Method overloading. If we have to perform only one operation, having same of the methods increases the readability of the program. Suppose you have to perform addition of the given numbers but there can be any number of arguments, if you write the method such as (int, int) for two parameters, and b(int, int, int) for three parameters then it may be difficult for you as well as other programmers to understand the behaviour of the method beacause its name differs. So we perform method overloading to figure out the program quickly.

Advantages of method Overloading :

Method overloading increases the readability of the program.

Different ways to overload the method :

There are two ways to overload the method in java :

1. By changing number of arguments
2. By changing the data type

Method Overloading : changing no. of arguments

In this example, we have created two methods

first add() method performs addition of two numbers and second add() method performs addition of three numbers.

```
class Adder{
static int add (int a, int b){return a+b;}
static int add(int a, int b, int c){return a+b+c;}
}
class TestOverloading1{
public static void main(String[] args){
  System.out.println(Adder.add(11,11));
  System.out.println(Adder.add(11,11,11));
}}
```

output:

22
33

Method Overloading: changing data type of arguments

In this example, we have created two methods that differs in data type. The first add method receives two integer arguments and second add method receives two double arguments.

```
class Adder{
static int add(int a, int b){return a+b;}
static double add(double a, double b){return a+b;}
}
class TestOverloading2{
public static void main(String[] args){
System.out.println(Adder.add(11,11));
System.out.println(Adder.add(12.3,12.6));
}}
```

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output:
22
24.9

7.2 Method Overriding in Java:

If subclass (child class) has the same method as declared in the parent class, it is known as method overriding in java. In other words, if a subclass provides the specific implementation of the method that has been declared by one of its parent class, it is known as method overriding.

Usage of Java Method Overriding:

- Method Overiding is used to provide the specific implementation of a method which is already provided by its superclass.
- Method Overriding is used for runtime polymorphism.

Rules for Java Method Overriding:

The method must have the same name as in the parent class.
The method must have the same parameter as in the parent class
There must be an IS-A relationship (inheritance).

Example of Method Overriding:

```
// Java program to illustrate the use of Java Method Overriding
// creating a parent class
class vehicle {
   // defining a method
```